

TS4 - Trends

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Contents

1	Nonstationary Time Series	5
1.1	Deterministic and Stochastic trends	5
1.2	Trends and Logarithms	6
1.3	Comparison: Deterministic vs. Stochastic trends	6
1.3.1	Forecasts	6
1.3.2	Variance of the Forecast Errors	7
1.3.3	Dynamic Multipliers	8
1.3.4	Removing the trend	9
1.4	Trends and real data	9
1.5	Spurious regression	10
1.6	Monte Carlo and Unit Roots	11
1.7	Dickey-Fuller tests	12
1.7.1	Pure Random Walk	12
1.7.2	Random Walk + Drift	14
1.7.3	Random Walk + Drift + Deterministic Trend	15
1.8	Augmented Dickey-Fuller tests	16
1.8.1	Example 1	17
1.8.2	Example 2	17
1.8.3	Example 3	18
1.9	Extensions of the Dickey-Fuller Test	18
1.9.1	Selection of the Lag Length	18
1.9.2	Multiple Roots	19
1.9.3	Seasonal Differencing	19
1.9.4	Structural Change	20
1.10	Deterministic Terms and Power of a Test	20
1.11	PP test	23
A	Appendix	25
A.1	Brownian Motion	25
A.2	Martingales	27
A.3	Functional Central Limit Theorem	29
A.4	Proof Theorem 1.1	30

Chapter 1

Nonstationary Time Series

1.1 Deterministic and Stochastic trends

There are two popular approaches to describe trends¹. The first one is to include a **deterministic time trend** and build the **Trend Stationary model**:

$$Y_t = c + \delta t + \Psi(L)\varepsilon_t, \quad (1.1)$$

The second one is to include a **stochastic trend** and specify a **Unit Root process**:

$$(1 - L)Y_t = \delta + \Psi(L)\varepsilon_t, \quad (1.2)$$

where $\varepsilon_t \sim w.n.$, $\Psi(L)\varepsilon_t \sim I(0)$ and $\Psi(L) = (1 + \psi_1 L + \psi_2 L^2 + \dots)$. Moreover, we assume $\Psi(1) \neq 0$ to rule out the possibility that Y_t is already stationary². To show this point, consider that $Y_t = \mu + \Gamma(L)\varepsilon_t$ is stationary. Applying $(1 - L)$ yields $(1 - L)Y_t = (1 - L)\Gamma(L)\varepsilon_t = \Psi(L)\varepsilon_t$, where $\Psi(L) = (1 - L)\Gamma(L)$ and $\Psi(1) = 0$. Furthermore, one can substitute recursively for Y_{t-1} in (1.2) to get

$$\begin{aligned} Y_t &= Y_{t-1} + \delta + \Psi(L)\varepsilon_t, \quad Y_0 = c \\ &= c + \delta t + \Psi(L) \sum_{i=1}^t \varepsilon_i \end{aligned} \quad (1.3)$$

One example of a unit root process is the **random walk with drift**, where $\Psi(L) = 1$:

$$Y_t = \delta + Y_{t-1} + \varepsilon_t, \quad Y_0 = c \quad (1.4)$$

$$Y_t = c + \delta t + \sum_{i=1}^t \varepsilon_i \quad (1.5)$$

A general way of representing a **Stochastic Trend model** for $d \geq 1$ is by the specification

$$Y_t = c + \delta t + u_t \quad (1.6)$$

where u_t is modeled as an ARIMA(p, d, q) process:

$$(1 - L)^d u_t = \frac{1 + \theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q}{(1 - \lambda_1 L)(1 - \lambda_2 L) \dots (1 - \lambda_p L)} \varepsilon_t = \Psi(L)\varepsilon_t, \quad (1.7)$$

¹A key feature of a trend is that it has a permanent effect on a series.

²In this case, differencing would introduce a unit root in the MA part of the process, making it non-invertible.

1.2 Trends and Logarithms

In a **Trend Stationary Model**, a **linear trend** δt , as in (1.1), implies that the growth rate $dY_t/dt = \delta$ is independent of the level of the series. However, most series are better characterized by an **exponential trend** $e^{\delta t}$, which leads to constant proportional growth $dY_t/dt = \delta Y_t$. An exponential trend is easily modelled by using the natural log of Y_t in (1.1):

$$\ln(Y_t) = c + \delta t + \Psi(L)\varepsilon_t \longrightarrow Y_t = \exp(c + \delta t + \Psi(L)\varepsilon_t) \longrightarrow dY_t/dt = \delta Y_t$$

Similar arguments suggest taking logs before applying differences in a **Stochastic Trend model**. For small changes, the first difference of the log of a variable is approximately the same as the percentage change in the variable

$$(1 - L)\ln(Y_t) = \ln(Y_t/Y_{t-1}) = \ln(1 + [(Y_t - Y_{t-1})/Y_{t-1}]) \approx (Y_t - Y_{t-1})/Y_{t-1}$$

since $\ln(1 + x) \approx x$. Therefore, if $\ln(Y_t)$ is a unit root process, its rate of growth is assumed stationary. The above developments explain why economists often model $\ln(GDP)$ instead of GDP in both cases of deterministic and/or stochastic trends.

1.3 Comparison: Deterministic vs. Stochastic trends

This section compares a trend-stationary process (1.1) with a unit root process (1.2) in terms of forecasts of the series, variance of the forecast error, dynamic multipliers, and transformations required to achieve stationarity.

1.3.1 Forecasts

The Forecasting of the **trend-stationary process** in (1.1) is given by:

$$Y_{t+s|t} = c + \delta(t + s) + \psi_s\varepsilon_t + \psi_{s+1}\varepsilon_{t-1} + \psi_{s+2}\varepsilon_{t-2} + \cdots \quad (1.8)$$

where $\varepsilon_{t+i|t} = 0 \forall i > 0$. The square summability of $\{\psi_j\}$ ³, i.e. $\sum_{j=0}^{\infty} \psi_j^2 < \infty$ implies that this forecast converges in mean square to the time trend:

$$E[(Y_{t+s|t} - c - \delta(t + s))^2] \rightarrow 0 \quad \text{as } s \rightarrow \infty. \quad (1.9)$$

For the **unit root process** in (1.2), ΔY_t can be forecasted using:

$$\begin{aligned} \Delta Y_{t+s|t} &\equiv E[(Y_{t+s} - Y_{t+s-1}) | \mathcal{F}_t] \\ &= \delta + \psi_s\varepsilon_t + \psi_{s+1}\varepsilon_{t-1} + \psi_{s+2}\varepsilon_{t-2} + \cdots \end{aligned} \quad (1.10)$$

Therefore,

$$\begin{aligned} Y_{t+s|t} &= \Delta Y_{t+s|t} + \Delta Y_{t+s-1|t} + \cdots + \Delta Y_{t+1|t} + Y_t \\ &= \{\delta + \psi_s\varepsilon_t + \psi_{s+1}\varepsilon_{t-1} + \psi_{s+2}\varepsilon_{t-2} + \cdots\} \\ &\quad + \{\delta + \psi_{s-1}\varepsilon_t + \psi_s\varepsilon_{t-1} + \psi_{s+1}\varepsilon_{t-2} + \cdots\} \\ &\quad + \cdots + \{\delta + \psi_1\varepsilon_t + \psi_2\varepsilon_{t-1} + \psi_3\varepsilon_{t-2} + \cdots\} + Y_t. \end{aligned}$$

and equivalently,

$$Y_{t+s|t} = s\delta + Y_t + (\psi_s + \psi_{s-1} + \cdots + \psi_1)\varepsilon_t + (\psi_{s+1} + \psi_s + \cdots + \psi_2)\varepsilon_{t-1} + \cdots. \quad (1.11)$$

³This condition ensures the process is stationary.

Substituting for the Y_t given in (1.3) yields

$$Y_{t+s|t} = (t+s)\delta + c + \Psi(L) \sum_{i=1}^t \varepsilon_i + (\psi_s + \psi_{s-1} + \cdots + \psi_1)\varepsilon_t + (\psi_{s+1} + \psi_s + \cdots + \psi_2)\varepsilon_{t-1} + \cdots. \quad (1.12)$$

For example, the **random walk with drift** (1.4), in which $\psi_0 = 1$ and $\psi_1 = \psi_2 = \cdots = 0$, results in

$$Y_{t+s|t} = s\delta + Y_t = (t+s)\delta + c + \sum_{i=1}^t \varepsilon_i \quad (1.13)$$

Analyzing (1.13), a random walk with drift δ is expected to grow at the constant rate of δ per period from whatever its current value Y_t happens to be. Equivalently, we can interpret that the model has a stochastic intercept term $c + \sum_{i=1}^t \varepsilon_i$.

Thus, the parameter δ in the unit root process (1.2) plays a similar role to that of δ in the deterministic time trend (1.1). With either specification, the forecast $Y_{t+s|t}$ in (1.8) or (1.11) converges to a linear function of the forecast horizon s with slope δ . The key difference is in the intercept of the line. For a trend-stationary process, the forecast converges to a line whose **deterministic intercept** is c , being the same regardless of the value of Y_t . By contrast, the **stochastic intercept** of the limiting forecast for a unit root process is given by $c + \Psi(L) \sum_{i=1}^t \varepsilon_i + (\psi_s + \psi_{s-1} + \cdots + \psi_1)\varepsilon_t + (\psi_{s+1} + \psi_s + \cdots + \psi_2)\varepsilon_{t-1} + \cdots$ in (1.12) and $c + \sum_{i=1}^t \varepsilon_i$ in the random walk with drift (1.13).

Therefore, the effect of a shock ε_t in the long-run mean $Y_{t+s|t}$ is temporary for the deterministic time trend (1.1) and permanent for the unit root process (1.2). Figure 1.1 displays the forecasts for these two types of trends.

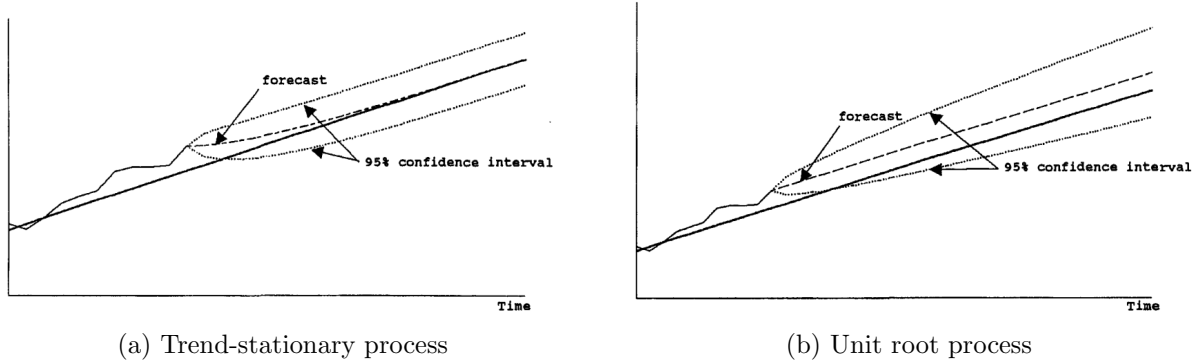


Figure 1.1: Forecast on TS series vs. Forecast on DS series.

1.3.2 Variance of the Forecast Errors

For the **trend-stationary process** (1.1), the **s-period-ahead forecast error** is:

$$\begin{aligned} Y_{t+s} - \hat{Y}_{t+s|t} &= \{c + \delta(t+s) + \varepsilon_{t+s} + \psi_1\varepsilon_{t+s-1} + \psi_2\varepsilon_{t+s-2} + \cdots \\ &\quad + \psi_{s-1}\varepsilon_{t+1} + \psi_s\varepsilon_t + \psi_{s+1}\varepsilon_{t-1} + \cdots\} \\ &\quad - \{c + \delta(t+s) + \psi_s\varepsilon_t + \psi_{s+1}\varepsilon_{t-1} + \psi_{s+2}\varepsilon_{t-2} + \cdots\} \\ &= \varepsilon_{t+s} + \psi_1\varepsilon_{t+s-1} + \psi_2\varepsilon_{t+s-2} + \cdots + \psi_{s-1}\varepsilon_{t+1} \end{aligned} \quad (1.14)$$

The **MSE** of this forecast is:

$$E[(Y_{t+s} - \hat{Y}_{t+s|t})^2] = (1 + \psi_1^2 + \psi_2^2 + \cdots + \psi_{s-1}^2)\sigma^2 \quad (1.15)$$

In the limit $s \rightarrow \infty$, this MSE becomes:

$$\lim_{s \rightarrow \infty} E[(Y_{t+s} - \hat{Y}_{t+s|t})^2] = (1 + \psi_1^2 + \psi_2^2 + \dots)\sigma^2$$

which is just the unconditional variance of the stationary component $\Psi(L)\varepsilon_t$, i.e. $\text{Var}[\Psi(L)\varepsilon_t]$. Therefore, as s becomes large, the added uncertainty from forecasting farther into the future becomes negligible

In contrast, for the **unit root process** (1.2), the **s-period-ahead forecast error** is:

$$\begin{aligned} Y_{t+s} - \hat{Y}_{t+s|t} &= \{\Delta Y_{t+s} + \Delta Y_{t+s-1} + \dots + \Delta Y_{t+1} + Y_t\} \\ &\quad - \{\Delta \hat{Y}_{t+s|t} + \Delta \hat{Y}_{t+s-1|t} + \dots + \Delta \hat{Y}_{t+1|t} + Y_t\} \\ &= \{\varepsilon_{t+s} + \psi_1 \varepsilon_{t+s-1} + \dots + \psi_{s-1} \varepsilon_{t+1}\} \\ &\quad + \{\varepsilon_{t+s-1} + \psi_1 \varepsilon_{t+s-2} + \dots + \psi_{s-2} \varepsilon_{t+1}\} + \dots + \{\varepsilon_{t+1}\} \\ &= \varepsilon_{t+s} + (1 + \psi_1) \varepsilon_{t+s-1} + (1 + \psi_1 + \psi_2) \varepsilon_{t+s-2} + \dots \\ &\quad + (1 + \psi_1 + \psi_2 + \dots + \psi_{s-1}) \varepsilon_{t+1} \end{aligned} \quad (1.16)$$

with **MSE**

$$E[(Y_{t+s} - \hat{Y}_{t+s|t})^2] = \{1 + (1 + \psi_1)^2 + (1 + \psi_1 + \psi_2)^2 + \dots + (1 + \psi_1 + \dots + \psi_{s-1})^2\} \sigma^2 \quad (1.17)$$

The MSE also increases with s , but does not converge to any fixed value as $s \rightarrow \infty$, since it grows linearly with slope $(1 + \psi_1 + \psi_2 + \dots)^2$. In the Random Walk, the MSE is $s\sigma^2$ and the slope is 1. Summarizing, for a trend-stationary process, the MSE reaches a finite bound as the forecast horizon becomes large, whereas for a unit root process the MSE eventually grows linearly with the forecast horizon.

One interest observation is the following: **(1)** The MSE grows linearly with the forecast horizon s , and thus the standard deviation of the forecast error grows with rate $\sqrt{s}$. However, **(2)** if $\delta > 0$, the forecast itself grows linearly in s , dominating $\sqrt{s}$. Therefore, the trend introduced by a nonzero drift δ *asymptotically dominates* the increasing variability arising over time due to the unit root component⁴. This is one of the main reasons why one should be cautious with stochastic trend models with drift. Nevertheless, the process remains difference-stationary rather than trend-stationary, since the variance around its deterministic trend increases without bound.

1.3.3 Dynamic Multipliers

Another difference between trend-stationary and unit root processes is the persistence of innovations. For trend-stationary processes, the dynamic multiplier is given by:

$$\frac{\partial Y_{t+s}}{\partial \varepsilon_t} = \psi_s, \quad (1.18)$$

and since $\lim_{s \rightarrow \infty} \frac{\partial Y_{t+s}}{\partial \varepsilon_t} = 0$, the effect of any stochastic disturbance eventually wears off. By contrast, for a unit root process, the dynamic multiplier is:

$$\frac{\partial Y_{t+s}}{\partial \varepsilon_t} = 1 + \psi_1 + \psi_2 + \dots + \psi_s, \quad (1.19)$$

where an innovation ε_t has a permanent effect on the level of Y_t . This is captured by

$$\lim_{s \rightarrow \infty} \frac{\partial Y_{t+s}}{\partial \varepsilon_t} = 1 + \psi_1 + \psi_2 + \dots = \psi(1). \quad (1.20)$$

⁴This implies that the data from a **unit root process with positive drift** are certain to exhibit an upward trend if observed for a sufficiently long period.

1.3.4 Removing the trend

A final difference between trend-stationary and unit root processes is the transformation required to achieve stationarity.

- **Stochastic trends:** Induce stationarity with **differencing**. This yields

$$\Delta Y_t = Y_t - Y_{t-1} = \delta + \Psi(L)\epsilon_t$$

In a deterministic trend, the first-difference returns:

$$\Delta Y_t = \delta + \Psi(L)(\epsilon_t - \epsilon_{t-1}) = \delta + \Psi(L)(1 - L)\epsilon_t$$

which is stationary process, but noninvertible due to a unit root in the MA part.

- **Deterministic trends:** Induce stationarity with **detrending**, by fitting a linear trend parametrically and removing it from the series

$$Y_t - c - \delta t = \Psi(L)(\epsilon_t)$$

For a stochastic trend, this does not result in stationarity, since:

$$Y_t - c - \delta t = \Psi(L) \sum_{i=1}^t \epsilon_i$$

remains unstationary. Moreover, one can detrend by fitting the series on a higher-order polynomial⁵. The degree of the polynomial can be determined by t-tests/F-tests, or information criteria (AIC, BIC). Simply subtracting the estimate trend values from Y_t in a trend-stationary process yields the stationary sequence $\{\Psi(L)(\epsilon_t)\}_{t=1}^T$. Then, the detrended process can be estimated by traditional methods, such as ARMA.

Visually, both **Trend Stationary (TS)** and **Difference Stationary (DS)** have a linear trend in the mean, but their variances are different. From an economic perspective, the difference is whether a shock (e.g. economic recessions) on the observed variable (e.g. GDP) is transitory (TS) or permanent (DS).

1.4 Trends and real data

The belief that trend is unchanging over time leads to the common practice of **detrending** macroeconomic data using a linear (or polynomial) deterministic regression equation. However, Nelson and Plosser [1982] challenged this traditional view, concluding that many macroeconomic variables tend to be **difference stationary (DS)** rather than **trend stationary (TS)**. They compared the ACFs of the original, 1st-differenced and detrended series. Note that the

	ρ_1	ρ_2	$\rho_{DS(1)}$	$\rho_{DS(2)}$	$\rho_{TS(1)}$	$\rho_{TS(2)}$
Real GNP	0.95	0.90	0.34	0.04	0.87	0.66
Nominal GNP	0.95	0.89	0.44	0.08	0.93	0.79
Industrial production	0.97	0.94	0.03	-0.11	0.84	0.67
Unemployment rate	0.75	0.47	0.09	-0.29	0.75	0.46

autocorrelations for the first 3 series indicate a unit root process. While the ACF for the **DS** series is stationary, the ACF for the **TS** series appears to still lead to a nonstationary process, being reasonably high. This is consistent with the fact that detrending a DS series will

⁵Alternatively, the trend may be estimated using a filtering method, e.g. HP filter.

not eliminate the nonstationarity. Moreover, none of the differenced series reported by Nelson and Plosser appears to have a unit root in the MA terms. Thus, **detrending** the above data does not result in a stationary series, while **differencing** does. The overall implication is that macroeconomic variables do not grow at a smooth long-run rate. Some macroeconomic shocks are of a permanent nature; the effects of such shocks are never eliminated.

Another important point is that one may be **cautious** when jointly modelling stochastic and deterministic trends, for example

$$Y_t = \delta t + Y_{t-1} + \varepsilon_t. \quad (1.21)$$

The deterministic time trend basically dominates the time path of the series. In a very large sample, asymptotic theory suggests this will always be the case. Furthermore, there is considerable estimation uncertainty, since OLS estimation it is difficult to distinguish between stochastic and deterministic trends.

1.5 Spurious regression

Consider the regression equation

$$Y_t = c + \phi Z_t + e_t \quad (1.22)$$

where e_t may be serially correlated. In the presence of **nonstationary variables**, (1.22) might be a **spurious regression**. A spurious regression has a high R^2 and t -statistics that appear to be significant, but the results have no economic meaning. The "fake" **OLS estimates are not consistent** and the **usual tests of statistical inference do not hold**. For example, consider two **independent** random walk sequences $\{Y_t\}$ and $\{Z_t\}$:

$$\begin{aligned} Y_t &= Y_{t-1} + \varepsilon_{yt}, \\ Z_t &= Z_{t-1} + \varepsilon_{zt}, \end{aligned}$$

in which $c = Y_0 = Z_0 = 0$. The residuals of this spurious regression are given by:

$$e_t = Y_t - \phi Z_t = \sum_{i=1}^t \varepsilon_{yi} - \phi \sum_{i=1}^t \varepsilon_{zi} = e_{t-1} + \varepsilon_{yt} - \phi \varepsilon_{zt}. \quad (1.23)$$

Thus, $\Delta e_t = \varepsilon_{yt} - \phi \varepsilon_{zt} \sim I(0)$, implying $e_t \sim I(1)$. Hence, the residual series $\{e_t\}$ contains a stochastic trend, violating the finite and stable moment conditions required for standard OLS asymptotics⁶:

1. $E(Z_t e_t \mid \mathcal{F}_{t-1}) = Z_{t-1} e_{t-1} - \phi \sigma_z^2 \neq 0$, so $\{Z_t e_t\}$ is not a martingale difference sequence and the usual martingale CLT does not apply;
2. $(1/T) \sum_{t=1}^T E(e_t^2 Z_t^2) \not\rightarrow \Omega < \infty$;
3. $(1/T) \sum_{t=1}^T Z_t^2 \not\rightarrow Q < \infty$.

Granger and Newbold [1974] simulated such an experiment and observed that, at the 5% level, ϕ was significantly $\neq 0$ at 75% of the time⁷. Since the OLS assumptions are violated, any t -test, F -test or R^2 values are unreliable. Furthermore, OLS is not consistent. In fact, one must make the series stationary to get any meaningful relationship.

There are four cases to consider when working with regressions:

⁶See Hamilton [1994] Chapter 7, Case 6.

⁷should be 5%.

- **Case 1:** Both $\{Y_t\}$ and $\{Z_t\}$ are **stationary**. Classical OLS regression is appropriate.
- **Case 2:** $\{Y_t\}$ and $\{Z_t\}$ are **integrated of different orders**. The regression is meaningless because no linear combination can eliminate the stochastic trend. For example, let $Y_t = Y_{t-1} + \varepsilon_{yt}$ and $Z_t = \rho Z_{t-1} + \varepsilon_{zt}$, with $|\rho| < 1$, so that $Y_t \sim I(1)$ and $Z_t \sim I(0)$. In the regression $Y_t = \phi Z_t + e_t$, we have $e_t = Y_t - \phi Z_t = \sum_{j=1}^t \varepsilon_{yj} - \phi \sum_{j=0}^{\infty} \rho^j \varepsilon_{z,t-j}$. Although $\sum_{j=0}^{\infty} \rho^j \varepsilon_{z,t-j}$ is stationary, e_t still contains the stochastic trend $\sum_{j=1}^t \varepsilon_{yj}$ and is therefore $I(1)$.
- **Case 3:** **Nonstationary** $\{Y_t\}$ and $\{Z_t\}$ **integrated of the same order**, and $\{e_t\}$ contains a stochastic trend. This is the case in which the regression is **spurious**, and this regression is meaningless. Differentiate to apply standard regression
- **Case 4:** **Nonstationary** $\{Y_t\}$ and $\{Z_t\}$ are **cointegrated**. A linear combination of the variables removes the stochastic trend.

1.6 Monte Carlo and Unit Roots

The **ACF** is not very reliable to verify unit roots, since near-unit root processes will have similar sample correlograms. Therefore, a formal procedure for testing on unit roots is needed. Consider the process:

$$Y_t = \phi Y_{t-1} + \varepsilon_t \quad (1.24)$$

When testing $H_0 : \phi = 0$ ⁸, OLS estimation and the standard asymptotic tests are valid under the null, since the process is stationary. However, when testing $H_0 : \phi = 1$, the sequence under H_0 is generated by the nonstationary process

$$Y_t = Y_0 + \sum_{i=1}^t \varepsilon_i,$$

which leads to a **spurious regression**. Thus, classical regression methods to test the significance of ϕ do not apply. Furthermore, the estimated model may mimic an AR(1) process with a near unit root; and the usual **t-statistics distributions** will not be applicable⁹.

However, if we knew the **true distribution of ϕ under $H_0 : \phi = 1$** , we could perform this significance test. Dickey and Fuller [1979, 1981] generated asymptotic (very large T) critical values for the t -statistics under $H_0 : \phi = 1$ using classical OLS to test for unit roots:

- 90% of $\hat{\phi}$: less than 2.58 standard errors from unity
- 95% of $\hat{\phi}$: less than 2.89 standard errors from unity
- 99% of $\hat{\phi}$: less than 3.51 standard errors from unity

For example, suppose $\phi = 0.9247$ and $\text{se}(\phi) = 0.037$. Then the t -statistic under H_0 is $(1 - 0.9247)/0.037 = 2.035$. In classical OLS, we would reject $H_0 : \phi = 1$ in a 5% level. However, using the Dickey Fuller critical values, H_0 is not rejected even at the 10% level.

Moreover, $\phi = -1$ would also enter the case of unit root. However, such cases are rare and we perform unilateral tests of positive unit root all the time.

Therefore, the test statistics under nonstationarity are nonstandard and cannot be analytically evaluated with traditional tools. Accordingly, **Monte Carlo simulation** can tabulate the distribution of the desired parameters and/or sample statistics by replicating the actual DGP for a given sample size T . It has 4 steps:

⁸More generally, any $|\phi| < 1$.

⁹Because the usual t -statistics are based on standard asymptotic results for OLS. In the case of unit roots we'll have another distribution.

1. Generate the $\{\varepsilon_t\}_{t=1}^T$ sequence, e.g., from $\varepsilon_t \stackrel{iid}{\sim} N(0, 1)$.
2. Under H_0 , construct the $\{Y_t\}_{t=1}^T$ sequence as a random walk,

$$Y_t = Y_{t-1} + \varepsilon_t,$$

for the desired sample size T and a given initial value Y_0 , e.g., $Y_0 = 0$. Notice that constants and deterministic trends are not included when simulating $\{Y_t\}_{t=1}^T$; they are included only in the ADF regression used for estimation.

3. Estimate the chosen ADF regression,

$$\Delta Y_t = c + \delta t + \phi^* Y_{t-1} + \sum_{i=1}^p \beta_i \Delta Y_{t-i} + \varepsilon_t,$$

including the desired deterministic terms and lagged differences, and store the t -statistic associated with $H_0 : \phi^* = 0$.

4. Replicate Steps 1,2,3 K times. Thus, for each replication, the parameters of interest and the critical values are tabulated. This allows obtaining confidence intervals.

Monte Carlo testing is specially used for **small-sample** testing, where asymptotics results are not valid. For a specif example of generating the Dickey-Fuller distribution with $T = 100$ under normality, 1 constant and no additional lags:

1. Generate 100 $\{\varepsilon_t\}$ from $\mathcal{N}(0, 1)$.
2. Generate $Y_t = Y_{t-1} + \varepsilon_t$. For Y_0 use $Y_0 = 0$.
3. Estimate $\Delta Y_t = c + \gamma Y_{t-1} + \varepsilon_t$ and obtain the t -statistics for the null $H_0 : \gamma = 0$, i.e. $\hat{\gamma}/\text{se}(\hat{\gamma})$. **Note that the data is generated under the null hypothesis of a unit root and estimated under the alternative hypothesis.**
4. Replicate steps 1-3 K times and tabulate the results.

It is difficult to distinguish nonstationarity due to a deterministic trend from that due to a unit root. Moreover, most unit-root tests have low power against highly persistent stationary alternatives (e.g., it may be difficult to reject $H_0 : \phi = 1$ when $\phi = 0.99$). Their test statistics generally have nonstandard distributions under the null, with critical values obtained from tabulated or simulated distributions rather than the conventional t distribution.

1.7 Dickey-Fuller tests

In this section we formally derive the **analytical** DF test for the pure Random Walk model ¹⁰. Then we extend the results for the drift and deterministic trend cases.

1.7.1 Pure Random Walk

Consider the stationary AR(1) model

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim wn(0, \sigma^2), \quad (1.25)$$

where $c = (1 - \phi)\mu_t$, $\mu_t = E(Y_t)$ and $|\phi| < 1$. If $\hat{\phi}_{MLE}$ denotes the MLE of ϕ , then for T observations:

$$\hat{\phi}_{MLE} \sim \mathcal{N}(\phi, (1 - \phi^2)/T). \quad (1.26)$$

¹⁰In the last section we have shown how to generate the **empirical** distribution by Monte Carlo simulation.

To test the null hypothesis $H_0 : \phi = \phi_0$ against the alternative $H_1 : \phi \neq \phi_0$, we use the statistic

$$\frac{\hat{\phi}_{MLE} - \phi_0}{\text{s.e.}(\hat{\phi}_{MLE})}, \quad (1.27)$$

which has an approximate Student- t distribution under H_0 if $|\phi_0| < 1$. Observe that the distribution in (1.26) can also be written as

$$\sqrt{T}(\hat{\phi}_{MLE} - \phi) \sim \mathcal{N}(0, (1 - \phi^2)), \quad (1.28)$$

so that $\hat{\phi}_{MLE} = O_p(T^{-1/2})$, that is, the convergence rate of the estimator is $1/\sqrt{T}$. However, when $\phi = 1$, the **asymptotic variance** above becomes zero, which means we have a **degenerate distribution**. If $\phi = 1$ in (1.25), we obtain the pure **Random Walk**¹¹:

$$Y_t = \sum_{s=1}^t \varepsilon_s \quad \Rightarrow \quad Y_t \sim N(0, \sigma^2 t).$$

In the case $\phi = 1$, the MLE asymptotic distribution becomes degenerate and the normal approximation in (1.26) no longer applies; therefore, we cannot use the t -distribution to perform hypothesis testing on:

$$H_0 : \phi = 1, \quad H_1 : \phi < 1. \quad (1.29)$$

Since the MLE and OLS estimators are asymptotically equivalent¹², we have

$$\hat{\phi}_{OLS} = \frac{\sum_{t=1}^T Y_{t-1} Y_t}{\sum_{t=1}^T Y_{t-1}^2} = \phi + \frac{\sum_{t=1}^T Y_{t-1} \varepsilon_t}{\sum_{t=1}^T Y_{t-1}^2}. \quad (1.30)$$

It follows that

$$\hat{\phi}_{OLS} - \phi = \frac{\sum_{t=1}^T Y_{t-1} \varepsilon_t}{\sum_{t=1}^T Y_{t-1}^2}. \quad (1.31)$$

To test (1.29), we must study the behavior of

$$\hat{\phi}_{OLS} - 1 = \frac{\sum_{t=1}^T Y_{t-1} \varepsilon_t}{\sum_{t=1}^T Y_{t-1}^2}. \quad (1.32)$$

The AR(1) in eq. (1.25) (with $c = 0$) can be equivalently written as:

$$\Delta Y_t = \phi^* Y_{t-1} + \varepsilon_t, \quad (1.33)$$

where $\phi^* = \phi - 1$ and its OLS estimator can be obtained by regressing ΔY_t on Y_{t-1} . Therefore, testing (1.29) is equivalent to testing

$$H_0 : \phi^* = 0, \quad H_1 : \phi^* < 0. \quad (1.34)$$

Theorem 1.1. Consider model (1.25) with $Y_0 = 0$ and assume $\varepsilon_t \sim i.i.d.(0, \sigma^2)$ ¹³. Then,

$$T(\hat{\phi}_{OLS} - 1) \xrightarrow{D} \frac{1}{2} \frac{W(1)^2 - 1}{\int_0^1 [W(r)]^2 dr}, \quad (1.35)$$

where $W(r)$ denotes the **standard Brownian motion**, i.e. $W(r) \sim N(0, r)$.

¹¹ Assuming $c = 0$, for simplicity.

¹² if $\varepsilon_t \sim N(0, \sigma^2)$ in the MLE

¹³ Not necessarily $\sim N$

In particular, $W(1)^2 \sim \chi^2(1)$, and since $P(\chi^2(1) < 1) = 0.68$, from (1.35) we have that the probability of the left-hand side being negative converges to 0.68 as $T \rightarrow \infty$. That is, even if we have a random walk ($\phi = 1$), by simulating many samples of such process, approximately 2/3 of them will yield $\hat{\phi}_{OLS}$ less than one. Therefore, $\hat{\phi}_{OLS}$ is **biased** under the random walk.

From (1.35), we also observe that the convergence rate of the estimator differs from the stationary case, since:

$$\hat{\phi}_{OLS} = O_p(T^{-1}). \quad (1.36)$$

To test hypothesis (1.29) or (1.34), we use the statistic

$$\hat{\tau} = \frac{\hat{\phi}_{OLS}^*}{\text{s.e.}(\hat{\phi}_{OLS}^*)}, \quad (1.37)$$

where $\hat{\phi}_{OLS}^* = \hat{\phi}_{OLS} - 1$. Moreover, the **standard error** is given by OLS theory:

$$\text{s.e.}(\hat{\phi}_{OLS}^*) = \frac{S}{\left(\sum_{t=1}^T Y_{t-1}^2\right)^{1/2}}, \quad (1.38)$$

where S^2 is the unbiased estimator of σ^2 in regression (1.33).

$$S^2 = \frac{1}{T-2} \sum_{t=2}^T (\Delta Y_t - \hat{\phi}_{OLS}^* Y_{t-1})^2, \quad (1.39)$$

Hence, statistic (1.37) can be written equivalently as

$$\hat{\tau} = \frac{\hat{\phi}_{OLS} - 1}{(S^2 / \sum_{t=1}^T Y_{t-1}^2)^{1/2}} = \frac{\sum_{t=1}^T Y_{t-1} \varepsilon_t}{S}. \quad (1.40)$$

For samples of size $n = 100$, the critical values for 1%, 5% and 10% significance are -2.60 , -1.95 , and -1.61 . For $n = 500$ the critical values are -2.58 , -1.95 , and -1.62 . We reject H_0 whenever $\hat{\tau}$ is smaller than the appropriate critical value.

1.7.2 Random Walk + Drift

Consider the drift term is different from zero, i.e. $c \neq 0$. In this case, equation (1.33) is written as:

$$\Delta Y_t = c + \phi^* Y_{t-1} + \varepsilon_t, \quad (1.41)$$

where the OLS estimators are obtained by regressing ΔY_t into 1 and Y_{t-1} . Moreover, the presence of c alters the asymptotic distribution of $\hat{\phi}_{OLS}$ and $\hat{\tau}$. Instead of (1.35) and (1.40), we'll have:

$$T(\hat{\phi}_{OLS} - 1) \xRightarrow{D} \frac{\frac{1}{2}([W(1)]^2 - 1) - W(1) \int_0^1 W(r) dr}{\int_0^1 [W(r)]^2 dr - \left(\int_0^1 W(r) dr\right)^2}, \quad (1.42)$$

and

$$\hat{\tau}_\mu \xRightarrow{D} \frac{\frac{1}{2}([W(1)]^2 - 1) - W(1) \int_0^1 W(r) dr}{2[\int_0^1 [W(r)]^2 dr - \left(\int_0^1 W(r) dr\right)^2]^{1/2}}, \quad (1.43)$$

where $\hat{\tau}_\mu$ is used instead of $\hat{\tau}$. The distribution of $\hat{\tau}_\mu$ deviates further from the normal than $\hat{\tau}$. For large samples, the critical values of $\hat{\tau}_\mu$ for 1% and 5% significance are -3.42 and -2.86 , respectively.

1.7.3 Random Walk + Drift + Deterministic Trend

The previous procedure can be extended to include a linear trend. Consider the process:

$$\Delta Y_t = c + \delta t + \phi^* Y_{t-1} + \varepsilon_t. \quad (1.44)$$

The statistic for testing $H_0 : \phi_1^* = 0$ is given by:

$$\hat{\tau}_\tau = \frac{\hat{\phi}_{OLS}^* - 1}{\text{e.p.}(\hat{\phi}_{OLS}^*)}, \quad (1.45)$$

whose limiting distribution is given below.

Theorem 1.2. *Under i.i.d. errors with zero mean and variance σ^2 ,*

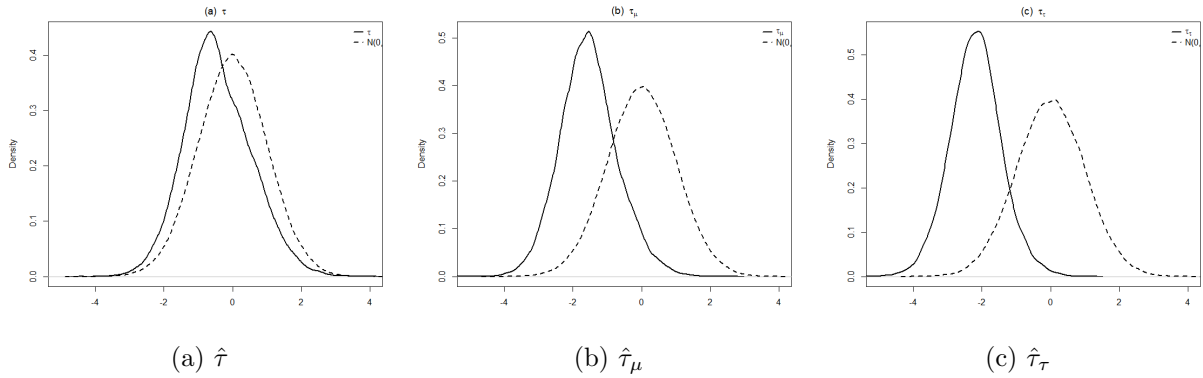
$$\hat{\tau}_\tau \xrightarrow{D} \frac{\frac{1}{2}([W(1)]^2 - 1) - W(1)\int_0^1 W(r) dr + A}{[\int_0^1 [W(r)]^2 dr - (\int_0^1 W(r) dr)^2 + B]^{1/2}}, \quad (1.46)$$

where

$$A = 12 \int_0^1 tW(t) dt - \frac{1}{2} \left(\int_0^1 W(t) dt \right)^2 - \frac{1}{2} W(1), \quad (1.47)$$

$$B = 12 \left(\int_0^1 tW(t) dt \right)^2 - \left(\int_0^1 tW(t) dt \right)^2 - 3 \int_0^1 W(t) dt^2. \quad (1.48)$$

The below figure shows the asymptotic distribution for $\hat{\tau}$, $\hat{\tau}_\mu$ and $\hat{\tau}_\tau$



Summarizing, Dickey and Fuller [1979] considered three different regression equations that can be used to test the presence of a unit root:

$$\Delta Y_t = \phi^* Y_{t-1} + \varepsilon_t. \quad (1.49)$$

$$\Delta Y_t = c + \phi^* Y_{t-1} + \varepsilon_t. \quad (1.50)$$

$$\Delta Y_t = c + \delta t + \phi^* Y_{t-1} + \varepsilon_t. \quad (1.51)$$

In all equations, if $\phi^* = 0$, the series contains a unit root. By estimating ϕ_{OLS}^* in a given regression setting and sample size¹⁴, we can perform t -tests using the appropriate values $\hat{\tau}/\hat{\tau}_\mu/\hat{\tau}_\tau$ reported in Dickey-Fuller tables. Additionally, Dickey and Fuller [1981] provide three additional F -statistics (called ϕ_1 , ϕ_2 and ϕ_3) to test **joint hypotheses** on the coefficients. The ϕ_1 , ϕ_2 and ϕ_3 statistics are constructed in exactly the same way as ordinary F -tests:

$$\phi_i = \frac{[\text{SSR}_r - \text{SSR}_{\text{ur}}]/r}{\text{SSR}_{\text{ur}}/(T - k)}. \quad (1.52)$$

¹⁴ Asymptotic critical values should be used only when the sample is sufficiently large; otherwise, finite-sample critical values appropriate to the regression specification and sample size should be used.

where SSR_r and SSR_{ur} correspond to the restricted ¹⁵ and unrestricted model ¹⁶; r is the number of restrictions and k is number of parameters estimated in the unrestricted model. Comparing the calculated value of ϕ_i to the appropriate value reported in Dickey and Fuller [1981] allows you to determine the significance level at which the restriction is binding. Since the null H_0 assumes the model is generated by the restricted model, a large value of ϕ_i implies the model is generated by the unrestricted model.

1.8 Augmented Dickey-Fuller tests

The above critical values $\hat{\tau}/\hat{\tau}_\mu/\hat{\tau}_\tau$ and $\phi_1/\phi_2/\phi_3$ are **asymptotically unchanged** if autoregressive components are included:

$$\Delta Y_t = \phi^* Y_{t-1} + \sum_{i=1}^p \beta_i \Delta Y_{t-i-1} + \varepsilon_t. \quad (1.53)$$

$$\Delta Y_t = c + \phi^* Y_{t-1} + \sum_{i=1}^p \beta_i \Delta Y_{t-i-1} + \varepsilon_t. \quad (1.54)$$

$$\Delta Y_t = c + \delta t + \phi^* Y_{t-1} + \sum_{i=1}^p \beta_i \Delta Y_{t-i-1} + \varepsilon_t. \quad (1.55)$$

Tests including lagged changes are called **Augmented Dickey–Fuller (ADF) tests**. Note that adding these extra differenced lags controls for serial autocorrelation in the errors, which makes the t and F -tests appropriate, but does not change the asymptotic critical values for the statistics. In contrast, adding too many lags may lead the **power** of the test to decrease. Nevertheless, the **finite-sample behavior** of the test changes, even though its **asymptotic distribution and critical values** remain the same. So be careful with the sample size when performing these tests.

Consider the general **AR**(p) model:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \varepsilon_t, \quad (1.56)$$

which can be written as

$$\Delta Y_t = c + \phi_1^* Y_{t-1} + \phi_2^* \Delta Y_{t-1} + \cdots + \phi_p^* \Delta Y_{t-p+1} + \varepsilon_t, \quad (1.57)$$

with $\phi_1^* = \sum_{i=1}^p \phi_i - 1$ and $\phi_j^* = -\sum_{i=j}^p \phi_i$ for $j \geq 2$. A unit root requires $\Phi(1) = 0$, i.e. $\sum_{i=1}^p \phi_i = 1$, hence $\phi_1^* = 0$. Testing for a unit root is therefore equivalent to testing $\phi_1^* = 0$. Recalling, the ADF statistics $T(\hat{\phi}_1^* - 1)$ and $\hat{\tau}_\mu = \hat{\phi}_1^*/\text{s.e.}(\hat{\phi}_1^*)$ share the same asymptotic distributions given in (1.42) and (1.43).

When $Y_t \sim \text{ARMA}(p, q)$, **Said and Dickey (1985)** showed that $\hat{\tau}_\mu$ computed from

$$\Delta Y_t = \theta_0 + \phi_1^* Y_{t-1} + \sum_{i=2}^p \phi_i^* \Delta Y_{t-i-1} - \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t, \quad (1.58)$$

has the same asymptotic distribution as in the model without the MA component.

Summarizing, for large T we can use the test statistics $\hat{\tau}/\hat{\tau}_\mu/\hat{\tau}_\tau$ and $\phi_1/\phi_2/\phi_3$ for all **ARMA**(p, q) models. **However, in small samples, these statistics can be different.** Additionally, estimating a large number of coefficients leads to reduced degrees of freedom and large standard errors, which consequently **lowers the power of the ADF test**.

¹⁵AR(p) in first-differences without $c/\delta/\phi^*$

¹⁶AR(p) in first-differences with $c/\delta/\phi^*$

1.8.1 Example 1

Dickey and Fuller (1981) analyze quarterly data for the logarithm of the Fed's Production Index (1950Q1–1977Q4) and estimate the following models:

$$\begin{aligned}\Delta y_t &= 0.52 + 0.00120t - 0.119y_{t-1} + 0.498\Delta y_{t-1} + \varepsilon_t, & \text{SSR} &= 0.056448, \\ \Delta y_t &= 0.0054 + 0.447\Delta y_{t-1} + \varepsilon_t, & \text{SSR} &= 0.063211, \\ \Delta y_t &= 0.511\Delta y_{t-1} + \varepsilon_t, & \text{SSR} &= 0.065966.\end{aligned}$$

Testing the **random-walk** specification (the restricted model) against the fully augmented model (the unrestricted one), they consider the null $H_0 : c = \phi^* = \gamma = 0$, which imposes 3 restrictions. With $T = 110$ and $k = 4$ estimated parameters, the unrestricted model has 106 degrees of freedom, so that $\text{SSR}_{ur}/106 = 0.000533$. The ϕ_2 statistic is therefore

$$\phi_2 = \frac{0.065966 - 0.056448}{3(0.000533)} = 5.95.$$

The critical value at the 2.5% level is 5.59, so they reject the random-walk null in favor of a model containing an intercept and/or a trend and/or a unit root.

They then test $\phi^* = \gamma = 0$ using the **random-walk plus drift** as the restricted model. The ϕ_3 statistic is

$$\phi_3 = \frac{0.063211 - 0.056448}{2(0.000533)} = 6.34.$$

With 110 observations, the 5% and 10% critical values are 6.49 and 5.47. Thus, they reject the null at 10% in favor of the full unrestricted model, but not at 5%.

Having selected the broader model using the F -tests, we may use the τ_τ test, yielding

$$\tau_\tau = -0.119/0.033 = -3.61,$$

which is significant at the 5% level, leading to rejection of the unit-root null. Therefore, it is recommended to apply unit-roots tests in 3 steps:

1. Select the appropriate lag-length p .
2. Use Dickey and Fuller F -tests or IC to select the best configuration (RW vs. RW + drift vs RW + drift + trend).
3. Perform Dickey and Fuller t -tests for the null of unit root on ϕ^* .

1.8.2 Example 2

Nelson and Plosser [1982] also performed formal testing of unit roots on real GNP (log), nominal GNP (log), industrial production (log) and unemployment. For each series they considered the ADF regression:

$$\Delta Y_t = c + \delta t + \phi^* Y_{t-1} + \sum_{i=1}^p \beta_i \Delta Y_{t-i-1} + \varepsilon_t.$$

with lags 2, 4, 6 respectively. They rejected the null of unit root, i.e. $\phi^* = 0$, only for unemployment.

1.8.3 Example 3

To test if the Long-run PPP holds, Enders [1988] checks if $r_t = e_t + p_t^* - p_t$ is stationary. They have considered the following regression without deterministic trends¹⁷:

$$\Delta r_t = c + \phi^* r_{t-1} + \sum_{i=1}^p \beta_i \Delta r_{t-i-1} + \varepsilon_t$$

for different countries and for different periods. All alternatives did not reject the null of unit root. This is mainly because unit root tests do not have much power in discriminating between characteristic roots close to unity and actual unit roots. Furthermore, PPP may be flawed.

1.9 Extensions of the Dickey-Fuller Test

The Dickey-Fuller test assumes independent and homoskedastic errors, raising several practical issues:

- The correct AR lag length may be unknown. If the regression is misspecified, ϕ^* and its standard error cannot be properly estimated¹⁸.
- The DGP may contain both AR and MA components, requiring a more general specification.
- The series may contain more than one unit root, i.e., $d > 1$. ADF tests are designed to test for a single unit root, so higher orders of integration require additional testing.
- Seasonal unit roots may be present, requiring tests specifically designed for seasonal integration.
- Structural breaks may mimic stochastic trends, leading to misspecification and potentially misleading unit-root tests.
- The appropriate deterministic terms (constant and/or trend) may be unknown, and their specification affects the test distribution and critical values.

1.9.1 Selection of the Lag Length

It is important to use the correct number of lags in the ADF test, since the conclusion may depend on p . Too few lags may leave serial correlation in the residuals and invalidate inference, whereas too many lags reduce the power of the ADF test.

A common approach is the **general-to-specific methodology**: start with a sufficiently large $p_{\max}$ and sequentially remove the last lag using standard t - or F -tests until the last included lag is significant. These standard tests are valid because the coefficients β_i multiply stationary variables ΔY_{t-i} , even when Y_{t-1} is $I(1)$ [Sims et al. [1990]]. Thus, in

$$\Delta Y_t = c + \phi^* Y_{t-1} + \sum_{i=1}^p \beta_i \Delta Y_{t-i} + \varepsilon_t,$$

standard t - and F -tests may be used for the β_i 's, whereas $H_0 : \phi^* = 0$ requires the Dickey-Fuller distribution. Information criteria may also be used to select p . In contrast, Monte Carlo studies show that the **specific-to-general methodology** is biased for a lower value of p . It is also possible to determine the lag length using **Information Criteria**, such as AIC or BIC,

¹⁷Not allowed in PPP theory.

¹⁸A similar issue arises when the DGP contains MA components.

as implemented in `urca::ur.df()`. It is also advisable to use appropriate t - and F -tests to determine whether a constant and/or trend should be included.

Once a tentative p and other regressors have been selected, perform **diagnostic tests** on the residuals to determine whether the regression specification needs to be modified¹⁹. In some cases, altering the regression specification requires critical values different from the usual Dickey-Fuller critical values. See Table 1.1.

Then, a **unit-root test** can be performed using the selected specification.

With respect to **MA components**, a stationary and invertible ARMA(p, q) process admits an AR(∞) representation and can therefore be approximated by an AR(p^*) model with sufficiently large p^* . Thus, explicit MA terms are not required in the ADF regression. The only problem is that this lowers the power of the test.

1.9.2 Multiple Roots

The most common method to determine the presence of multiple unit roots is the **bottom-up approach** that sequentially differentiates the series and test for the presence of an unit root. This means testing for a unit root, and then, if H_0 cannot be rejected, test for the presence of a second root. However, in repeated samples, this method is biased to select fewer roots. The **top-down approach** of Dickey and Pantula [1987] is more advisable for $I(2)$ and higher order series.

The methodology consists of performing Dickey-Fuller tests on successive differences of $\{Y_t\}$. For example, if at most two unit roots are suspected, the **Dickey-Pantula procedure** starts by testing for a second unit root from

$$\Delta^2 Y_t = c + \beta_1 \Delta Y_{t-1} + \varepsilon_t.$$

If $H_0 : \beta_1 = 0$ is not rejected, ΔY_t contains a unit root and hence $Y_t \sim I(2)$. Otherwise, $I(2)$ is rejected and we proceed to test for a single unit root from

$$\Delta^2 Y_t = c + \beta_1 \Delta Y_{t-1} + \beta_2 Y_{t-1} + \varepsilon_t.$$

To see the role of β_2 , consider the underlying AR(2), $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$. Rewriting it in the form above gives

$$\beta_1 = -(1 + \phi_2), \quad \beta_2 = \phi_1 + \phi_2 - 1.$$

Since the AR polynomial is $\Phi(L) = 1 - \phi_1 L - \phi_2 L^2$, a unit root requires $\Phi(1) = 1 - \phi_1 - \phi_2 = 0$, which is equivalent to $\beta_2 = 0$. Thus, under $H_0 : \beta_2 = 0$, one unit root remains and, since $I(2)$ has already been rejected, ΔY_t is stationary and $Y_t \sim I(1)$. Under $H_1 : \beta_2 < 0$, the remaining unit root is rejected, implying $Y_t \sim I(0)$. The appropriate Dickey-Fuller critical values must be used.

1.9.3 Seasonal Differencing

Seasonality can be avoided by working with a **seasonally adjusted series**, such as one obtained from X-13ARIMA-SEATS. In this case, seasonal differencing is generally unnecessary, although residual seasonality should still be checked. Particular care is required in forecasting exercises, since seasonal adjustment is revised as new observations become available. To avoid **data leakage**, real-time exercises should therefore use the **data vintage** available at each forecast origin.

¹⁹E.g., by adding further lags to remove serial correlation, deterministic regressors to account for structural changes, or dummy variables for outliers. Residual heteroskedasticity and nonnormality should also be examined, which affect finite-sample properties and power.

When working with the **raw series**, nonstationarity may arise from a regular unit root, associated with a stochastic trend, and/or from **seasonal unit roots**. Deterministic seasonality does not require seasonal differencing and can instead be controlled by seasonal dummies. For example, the ADF regression can be augmented as

$$\Delta Y_t = c + \phi^* Y_{t-1} + \sum_{j=1}^{s-1} \delta_j D_{jt} + \sum_{i=1}^p \beta_i \Delta Y_{t-i} + \varepsilon_t,$$

where D_{jt} are seasonal dummies. The hypothesis $H_0 : \phi^* = 0$ continues to test for a unit root at the zero frequency, as the inclusion of deterministic seasonal dummies does not alter the asymptotic Dickey–Fuller distribution of the test statistic.

If seasonality is stochastic, **seasonal unit-root tests** should be considered. Tests such as OCSB can be used to assess whether a seasonal difference

$$\Delta_s Y_t = (1 - L^s) Y_t$$

is required, while the **HEGY test** decomposes seasonal unit roots across the different seasonal frequencies and can identify which seasonal roots are present.

1.9.4 Structural Change

Structural changes may cause Dickey–Fuller tests to be biased toward the nonrejection of a unit root. For example, consider a stationary AR(1) process subject to a permanent change in its mean:

$$Y_t = c + \delta D_t + \phi Y_{t-1} + \varepsilon_t, \quad |\phi| < 1,$$

where $D_t = 0$ before the break and $D_t = 1$ afterward. Although $\{Y_t\}$ is stationary around different means within each regime, omitting D_t makes observations before and after the break appear highly persistent: low values tend to be followed by low values and high values by high values. Consequently, the estimated autoregressive coefficient is biased toward unity, making the Dickey–Fuller test less likely to reject the unit-root null.

If the break date is known, the natural solution is to augment the ADF regression with dummy variables that control for the structural change. For example, allowing for a break in the level,

$$\Delta Y_t = c + \phi^* Y_{t-1} + \delta_1 DP_t + \delta_2 DL_t + \sum_{i=1}^p \beta_i \Delta Y_{t-i} + \varepsilon_t,$$

where DP_t is a pulse dummy for the break period and DL_t is a level dummy for the post-break period. However, the usual Dickey–Fuller critical values are no longer appropriate. In this case, the **Perron test** should be used, which essentially performs the unit-root test controlling for a known structural break and provides the appropriate critical values.

We now summarize the asymptotic distribution of the ADF test discussed so far for different regression components in Table 1.1.

1.10 Deterministic Terms and Power of a Test

Unit root tests have low power²⁰ to distinguish a near-unit-root process from a true unit-root process. A simple Monte Carlo study summarized in Table 1.2 reports the proportion of rejections of the false unit-root null based on 10,000 simulations of the model $y_t = c + a_1 y_{t-1} + u_t$, with different values of $a_1 < 1$.

²⁰The power of a test is the probability of rejecting a false null hypothesis, i.e., one minus the probability of a Type II error.

Table 1.1: Asymptotic distribution of the ADF test under different components

Component	Changes ADF distribution($T \rightarrow \infty$)?	Treatment
AR dynamics	No	Add lags of ΔY_t
Invertible MA dynamics	No	Sufficient AR augmentation
Intercept	Yes	DF with intercept
Deterministic trend	Yes	DF with trend
Seasonal deterministic dummies	No	Include seasonal dummies
Seasonal stochastic trend	Yes	Seasonal unit-root tests (e.g., HEGY)
Higher-order stochastic trend, $I(d)$	Yes	Dickey–Pantula procedure
Structural break	Yes	Perron-type tests

Table 1.2: Monte Carlo rejection frequencies of the unit-root null

a_1	10%	5%	1%
0.80	95.9	87.4	51.4
0.90	52.1	33.1	9.0
0.95	23.4	12.7	2.6
0.99	10.5	5.8	1.3

However, in the case where a highly persistent stationary series is differenced, not all is lost. There is a trade-off: differencing a stationary series introduces misspecification, but estimating a root close to unity involves substantial estimation uncertainty. When the root is sufficiently close to one, the cost of this misspecification may be smaller than the gain from avoiding estimation uncertainty, so the differenced model may even provide better short-run forecasts.

Furthermore, the power of unit root tests is intrinsically related to the inclusion of deterministic regressors (constant and trend).

- Including unnecessary deterministic regressors, such as an intercept or trend, requires estimating additional parameters and generally reduces the power of the test. Thus, it is not a good idea to automatically estimate the most general specification with an intercept, trend, and many lags.
- However, incorrectly omitting a deterministic component that is present in the data-generating process, such as an intercept or time trend, can severely reduce the power of the test and may even cause it to approach zero.

In cases of complete uncertainty about the deterministic regressors²¹, there is a simple sequential procedure that starts from the most general specification and progressively removes unnecessary deterministic terms. Consider the regression equation

$$\Delta Y_t = c + \delta t + \phi^* Y_{t-1} + \sum_{i=1}^p \beta_i \Delta Y_{t-i} + \varepsilon_t,$$

1. Estimate the ADF regression with δ and c . Test $H_0 : \phi^* = 0$ using τ_τ .

(a) $H_0 : \phi^* = 0$ rejected: process does not contain a unit root²².

(b) $H_0 : \phi^* = 0$ not rejected: test $H_0 : \phi^* = \delta = 0$ with the ϕ_3 statistic.

i. $H_0 : \phi^* = \delta = 0$ not rejected: assume $\delta = 0$ and proceed to **Step 2**.

ii. $H_0 : \phi^* = \delta = 0$ rejected: estimate $\Delta Y_t = c + \delta t + \sum_{i=1}^p \beta_i \Delta Y_{t-i} + \varepsilon_t$ and perform a t -test on δ :

A. $H_0 : \delta = 0$ not rejected: remove the trend and proceed to **Step 2**.

²¹For example, PPP does not admit a deterministic trend.

²²This specification yields the test with the lowest power.

- B. $H_0 : \delta = 0$ rejected: conclude the process contains a deterministic trend. Reassess $H_0 : \phi^* = 0$ in the unrestricted ADF regression using the standard t distribution²³:
 - $H_0 : \phi^* = 0$ rejected: process is trend stationary (TS).
 - $H_0 : \phi^* = 0$ not rejected: process contains a deterministic and a stochastic trend (TS+DS).
2. Estimate the ADF regression with c . Test $H_0 : \phi^* = 0$ using τ_μ .
 - (a) $H_0 : \phi^* = 0$ rejected: process does not contain a unit root.
 - (b) $H_0 : \phi^* = 0$ not rejected: test $H_0 : \phi^* = c = 0$ with the ϕ_1 statistic.
 - i. $H_0 : \phi^* = c = 0$ not rejected: proceed to **Step 3**.
 - ii. $H_0 : \phi^* = c = 0$ rejected: estimate $\Delta Y_t = c + \sum_{i=1}^p \beta_i \Delta Y_{t-i} + \varepsilon_t$ and perform a t -test on c .
 - A. $H_0 : c = 0$ not rejected: proceed to **Step 3**.
 - B. $H_0 : c = 0$ rejected: conclude that the process contains an intercept. Reassess $H_0 : \phi^* = 0$ in the unrestricted ADF regression using the standard t distribution²⁴:
 - $H_0 : \phi^* = 0$ rejected: conclude stationarity around a nonzero mean.
 - $H_0 : \phi^* = 0$ not rejected: conclude a unit root with drift.
3. Finally, if neither δ and c are required, estimate the ADF regression without deterministic terms and test $H_0 : \phi^* = 0$ using τ .

We now state an important theorem used in the above procedure to determine when conventional inference is valid in the presence of integrated variables and deterministic regressors.

Theorem 1.3 (Sims et al. [1990]). *Suppose that the DGP contains deterministic regressors, such as an intercept or a time trend, and that these deterministic regressors are correctly included in the estimated regression. Then conventional asymptotic inference using t -tests or F -tests is valid for tests on the regression coefficients, even when the regression contains integrated variables.*

The result follows because, when a restriction involves parameters with different rates of convergence, its asymptotic distribution is dominated by the parameters with the slowest rate of convergence. Consequently, the presence and correct inclusion of the deterministic regressors restores conventional asymptotic inference for such restrictions.

Notice the distinction between simulating the ADF null distribution and applying Theorem 1.3. To construct the standard ADF distribution, Y_t is simulated as a pure random walk and deterministic regressors are included only in the estimating equation. Hence, Theorem 1.3 does not apply, and the resulting t -statistic has a nonstandard Dickey–Fuller distribution. In contrast, if a deterministic component is actually present in the DGP and is also included in the estimating equation, the conditions of Theorem 1.3 are satisfied and conventional t - and F -inference can be used.

Nevertheless, this methodology is subject to some limitations. The supplementary material of Enders [1995] applies this procedure to $\log GDP$ and concludes that the series is stationary with drift. Well, this does not make sense. It happens mainly because errors in low power testing from early stages propagate to subsequent stages.

Notice that c and δ have different roles under H_0 and H_1 : under $H_0 : \phi^* = 0$, c generates a linear trend and δ a quadratic trend in Y_t , whereas under $H_1 : \phi^* < 0$, c determines the level and

²³Theorem 1.3

²⁴Theorem 1.3

δ the linear trend. Furthermore, deterministic trends are often poorly estimated in the presence of a unit or near-unit root, since least squares has difficulty distinguishing deterministic trend movements from highly persistent stochastic movements. Enders [1995] explains two additional tests with higher power than the ADF test:

1. Schmidt et al. [1992]: LM-type unit-root tests designed to account for deterministic trends without jointly estimating them with the autoregressive parameter.
2. Elliott et al. [1996]: DF-GLS tests that first remove the deterministic components by GLS and then apply an ADF-type test.

The higher power of these tests arises from pre-estimating and removing the deterministic components, thus avoiding their joint estimation with the unit-root parameter. More flexible detrending may capture the "true" deterministic behavior, but it changes the unit-root test distribution. Moreover, if overly flexible, it may absorb part of the stochastic trend and leave you with bananas.

1.11 PP test

Phillips and Perron [1988] developed a unit-root test robust to residual autocorrelation and heteroskedasticity by using robust standard errors to modify the Dickey-Fuller test statistics. Unlike the ADF test, which controls for serial correlation through lagged differences, the Phillips–Perron test uses test statistics that are robust to autocorrelation and heteroskedasticity. The model is given by

$$\begin{aligned}\Delta Y_t &= c + \phi^* Y_{t-1} + \varepsilon_t, \\ \varepsilon_t &\sim I(0),\end{aligned}$$

and it is considered under appropriate regularity conditions. The DF statistics are modified using robust standard errors to account for autocorrelation and heteroskedasticity in ε_t .

Appendix A

Appendix

A.1 Brownian Motion

Definition A.1. The *Standard Brownian Motion (SBM)*¹ is a continuous process $W = \{W(t), 0 \leq t \leq 1\}$ that satisfies:

1. $W(0) = 0$.
2. $W(t_2) - W(t_1), W(t_3) - W(t_2), \dots, W(t_k) - W(t_{k-1})$ are independent.
3. $W(t) - W(s)$ and $W(t + \tau) - W(s + \tau)$ are identically distributed².
4. $W(t) \sim N(0, t), \quad \forall 0 \leq t \leq 1$.
5. The trajectories of $W(t)$, i.e. $\{W(t)\}_{t=0}^1$, are continuous with probability one.

From property **2**, the **SBM** has independent increments, from property **3** it has strictly stationary increments and from property **4** the **SBM** is a Gaussian process. Additionally, properties **1**, **3** and **4** imply that $W(t - s) \stackrel{d}{=} W(t) - W(s)$ with distribution $\mathcal{N}(0, t - s)$

$$W(t) - W(s) \stackrel{d}{=} W(t - s) - W(0) \stackrel{d}{=} W(t - s), \quad (\text{A.1})$$

$$W(t - s) \sim \mathcal{N}(0, t - s), \quad (\text{A.2})$$

for $s < t$. From property **4**, the **SBM** has **mean** $E(W(t)) = 0$ and **variance** $\text{Var}(W(t)) = t$, for $0 \leq t \leq 1$. To compute the **autocovariance function** of W , consider $s < t$. Then,

$$\begin{aligned} \gamma(t, s) &= E(W(t)W(s)) \\ &= E\{[(W(t) - W(s)) + W(s)]W(s)\} \\ &= E\{(W(t) - W(s))W(s)\} + E(W(s)^2). \end{aligned}$$

Since $W(t) - W(s)$ and $W(s) - W(0) = W(s)$ are independent, it follows that

$$\begin{aligned} \gamma(t, s) &= E(W(t) - W(s))E(W(s)) + s \\ &= 0 + s = s, \quad 0 \leq s < t. \end{aligned}$$

Similarly, if $t < s$, then $\gamma(t, s) = t$. Therefore,

$$\gamma(t, s) = \min(t, s). \quad (\text{A.3})$$

Since every **Gaussian process** is characterized by its mean and covariance, an **equivalent definition** of the **SBM** is the following:

¹Winer Process.

²And independent, implicit from property **2**

Definition A.2. A Standard Brownian Motion W is a Gaussian process with mean $E(W(t)) = 0$ and autocovariance $\gamma(t, s) = \min(t, s)$.

Although the trajectories of W are continuous, the fact that it has independent increments makes them quite irregular. In reality, a trajectory of an **SBM** is an abstract mathematical object that does not possess a real graphical representation, due to the following Facts 1 and 2. Fact 3 only displays a property for SBMs:

Fact 1. The trajectories of an SBM are not differentiable at any point.

Fact 2. The trajectories of an SBM do not have bounded variation on any finite interval.³

Fact 3. The SBM is 0.5-self-similar, that is,

$$(\tau^{1/2}W(t_1), \dots, \tau^{1/2}W(t_n)) \stackrel{d}{=} (W(\tau t_1), \dots, W(\tau t_n)),$$

for all τ and for any $t_1, \dots, t_n$ in the interval $[0, 1]$, $n \geq 1$.

Definition A.3. A stochastic process $X = \{X(t), t \geq 0\}$ is called **f-self-similar**, for some $f > 0$, if its finite-dimensional distributions satisfy

$$(\tau^f X(t_1), \dots, \tau^f X(t_n)) \stackrel{d}{=} (X(\tau t_1), \dots, X(\tau t_n)),$$

for all $\tau > 0$ and for any $t_1, \dots, t_n \geq 0$, $n \geq 1$. This property means that scaled parts of a trajectory over any interval have similar (but not identical) shapes.

Definition A.4. Consider the process $X(t) = \sigma W(t)$. Then, $X(t)$ has independent increments and $X(t) \sim N(0, \sigma^2 t)$. We can also consider the Brownian motion

$$X(t) = \mu t + \sigma W(t), \quad t \geq 0, \quad \sigma > 0, \quad \mu \in \mathbb{R}, \quad (\text{A.4})$$

which is called the **General Brownian motion**. From eq. (A.4), the process $X(t)$ is a Gaussian process with $E(X(t)) = \mu t$ and $\gamma_X(t, s) = \sigma^2 \min(t, s)$, for $s, t \geq 0$. The parameter μ is called the drift, while σ is called the volatility.

Definition A.5. For non-negative stochastic processes, e.g. prices, one defines the **Geometric Brownian Motion**:

$$X(t) = e^{\mu t + \sigma W(t)}, \quad t \geq 0,$$

where $\log\{X(t)\}$ is a General Brownian Motion. Since $\log(X(t)) \sim \mathcal{N}(\mu t, \sigma^2 t)$, then $X_t \sim \mathcal{LN}(\mu t, \sigma^2 t)$ and its mean and covariance are given by:

$$\begin{aligned} E(X(t)) &= e^{(\mu + 0.5\sigma^2)t}, \\ \gamma_X(t, s) &= e^{(\mu + 0.5\sigma^2)(t+s)} (e^{\sigma^2 s} - 1). \end{aligned}$$

for $0 \leq s \leq t$.

³Take a Stochastic Process class for better understanding this.

A.2 Martingales

A martingale is a stochastic process that represents the idea of a *fair game*. Suppose that X_n is the fortune of a player after the n -th game and let $\mathcal{F}_n$ denote the information available up to that time, i.e. the history $\mathcal{F}_n = \{X_1, \dots, X_n\}$. If the expected gain at the $(n+1)$ -th game, given the past information, is equal to the current fortune X_n , we say that the process $\{X_n, n \geq 1\}$ is a martingale.

Formally, let $X = \{X_t, t \in \mathcal{T}\}$, with $\mathcal{T} \subset \mathbb{R}$. For each $t \in \mathcal{T}$, X_t is a random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $\{\mathcal{F}_t, t \in \mathcal{T}\}$ is a non-decreasing family of sub- σ -algebras of $\mathcal{F}$, i.e. $\mathcal{F}_s \subset \mathcal{F}_t$ if $s \leq t$, and that X_t is $\mathcal{F}_t$ -measurable (adapted).

Definition A.6. A process $X = \{X_t, \mathcal{F}_t, t \in \mathcal{T}\}$ is a martingale if:

- (a) X_t is integrable for each $t \in \mathcal{T}$,
- (b) For $s \leq t$, we have

$$E(X_t | \mathcal{F}_s) = X_s. \quad (\text{A.5})$$

Condition (b) means that the best predictor of the future X_t , given the information up to time s , is simply the present value X_s . If instead $E(X_t | \mathcal{F}_s) \geq X_s$, the process is called a **sub-martingale**, and if $E(X_t | \mathcal{F}_s) \leq X_s$, it is a **super-martingale**.

In the discrete-time case, with $\mathcal{T} = \mathbb{Z}$ or $\mathcal{T} = \{1, 2, \dots\}$, the martingale condition becomes

$$E(X_{n+1} | \mathcal{F}_n) = X_n, \quad n \geq 1. \quad (\text{A.6})$$

Examples

- (a) If $X_1, X_2, \dots$ are independent random variables with $E(X_i) = 0$, then the partial sums $Y_n = \sum_{i=1}^n X_i$ form a martingale, since $E(Y_{n+1} | \mathcal{F}_n) = E(X_{n+1}) + Y_n = Y_n$.
- (b) With the same X_i as above, but $E(X_i) = \mu_i \geq 0$, the sums Y_n form a sub-martingale. If we center them as $Z_n = \sum_{i=1}^n (X_i - \mu_i)$, then Z_n is a martingale.
- (c) If $X_{n+1} = X_n + a_n$, $a_n \sim RB(0, \sigma^2)$, then $\{X_n, n \geq 1\}$ is a martingale. In other words, a **random walk** is a special case of a martingale. From the definition (A.6), we have

$$E(X_{n+1} - X_n | \mathcal{F}_n) = 0, \quad n \geq 1,$$

which we can write as

$$X_{n+1} = X_n + u_n,$$

where u_n need not be white noise; it may have constant variance or even be stationary.

Properties

1. If X_t and Y_t are martingales, then $X_t + Y_t$ will also be a martingale if both processes are martingales with respect to the same sequence of σ -algebras.
2. To verify that $\{X_n, \mathcal{F}_n, n \geq 1\}$ is a martingale, it suffices to show that condition (A.6) holds, if $\{X_n, \mathcal{F}_n, n \geq 1\}$ is such that $\mathcal{F}_n$ is an increasing family of sub- σ -algebras of $\mathcal{F}$, X_n is adapted to $\mathcal{F}_n$, and X_n is integrable for all $n \geq 1$.
3. If $\{X_n, n \geq 1\}$ is a martingale, then $\mathbb{E}(X_n) = \mathbb{E}(X_1)$ for all n .

4. If φ is a convex function and $\{X_n, \mathcal{F}_n, n \geq 1\}$ is a martingale with $\varphi(X_n)$ integrable for all $n \geq 1$, then $\{\varphi(X_n), \mathcal{F}_n, n \geq 1\}$ is a submartingale. Furthermore, if φ is also increasing and X_n is a submartingale, then $\varphi(X_n)$ will also be a submartingale.

The following result is a convergence theorem for submartingales.

Theorem A.1 (Doob). *Suppose that $\{X_n, \mathcal{F}_n, n \geq 1\}$ is a sub-martingale with $\sup_n E|X_n| < \infty$ (i.e. bounded in L^1 ⁴). Then $\{X_n\}$ converges almost surely to a limit X_∞ , which is integrable.*

Definition A.7. Let $\{Y_n, n \geq 1\}$ be a **zero-mean** stochastic process. If

$$E(Y_n | \mathcal{F}_{n-1}) = 0, \quad n \geq 2, \quad (\text{A.7})$$

then $\{Y_n\}$ is called a **martingale difference sequence** with respect to $\mathcal{F}_n$.

The **martingale difference** property is stronger than being **uncorrelated**. If $E[X_t | \mathcal{F}_{t-1}] = 0$, then for any random variable $Z \in \mathcal{F}_{t-1}$,

$$\mathbb{E}[X_t Z] = \mathbb{E}[\mathbb{E}[X_t Z | \mathcal{F}_{t-1}]] = \mathbb{E}[Z \mathbb{E}[X_t | \mathcal{F}_{t-1}]] = 0.$$

In particular, if we take $Z = X_{t-k}$ for any $k \geq 1$, we obtain

$$E[X_t X_{t-k}] = 0.$$

Thus, a martingale difference sequence is automatically uncorrelated over time. Therefore, the martingale difference property implies zero autocorrelation, which is stronger than merely requiring $\mathbb{E}[X_t X_{t-k}] = 0$ without conditioning.

However, a **martingale difference** is weaker than **independence**. Independence would require that X_t is independent of the entire past $\mathcal{F}_{t-1}$. The martingale difference property, however, only requires that

$$\mathbb{E}[X_t | \mathcal{F}_{t-1}] = 0, \quad (\text{A.8})$$

which allows X_t to still depend on past information through higher moments. For example, in a GARCH(1,1) model the innovations form a martingale difference sequence: they satisfy (A.8), but their conditional variance depends on $\mathcal{F}_{t-1}$. Thus, the sequence is not independent, since the distribution of X_t given the past is not the same as its unconditional distribution. Therefore, the martingale difference property is weaker than independence.

The following theorem provides a **CLT** equivalent for **martingale differences**:

Theorem A.2 (White). *Suppose that $\{Y_n, n \geq 1\}$ is a martingale difference sequence, and let*

$$\bar{Y}_N = \frac{1}{N} \sum_{n=1}^N Y_n.$$

If the following conditions hold:

$$(a) \ E(Y_n^2) = \sigma_n^2 > 0, \text{ with } \sum_{n=1}^N \sigma_n^2 / N \rightarrow \sigma^2 > 0,$$

$$(b) \ E|Y_n|^r < \infty \text{ for some } r > 2 \text{ and all } n,$$

$$(c) \ \sum_{n=1}^N Y_n^2 / N \xrightarrow{P} \sigma^2,$$

then

$$N^{1/2} \bar{Y}_N \xrightarrow{D} N(0, \sigma^2).$$

This is a **central limit theorem** for **averages of martingale differences**.

⁴The space L^1 is the set of integrable random variables (that is, $\mathbb{E}[|X|] < \infty$). Being **bounded in L^1** means that the entire sequence $\{X_n\}$ is not only integrable, but also that the expected absolute values do not grow indefinitely, that is, $\sup_{n \geq 1} \mathbb{E}[|X_n|] < \infty$.

A.3 Functional Central Limit Theorem

Suppose $Y_1, Y_2, \dots$ is a sequence of i.i.d. random variables with mean μ and variance σ^2 . Given the sample mean $\bar{Y}_N = \frac{1}{N} \sum_{t=1}^N Y_t$, the **usual Central Limit Theorem** affirms that:

$$\sqrt{N}(\bar{Y}_N - \mu) \xrightarrow{D} \mathcal{N}(0, \sigma^2). \quad (\text{A.9})$$

Moreover, the same holds for any proportion $0 \leq r \leq 1$ of the data $\bar{Y}_{Nr}$:

$$\sqrt{Nr}(\bar{Y}_{Nr} - \mu) \xrightarrow{D} \mathcal{N}(0, \sigma^2). \quad (\text{A.10})$$

In general, let's define

$$Y_N(r) = \frac{1}{N} \sum_{t=1}^{[Nr]} Y_t, \quad 0 \leq r \leq 1, \quad (\text{A.11})$$

which is proportional to the average of the first 100r% of the observations, and $Y_N(r) \neq \bar{Y}_{Nr}$ due to the denominator. We can also write

$$\sqrt{N}Y_N(r) = \frac{1}{\sqrt{N}} \sum_{t=1}^{[Nr]} Y_t = \frac{\sqrt{[Nr]}}{\sqrt{N}} \cdot \frac{1}{\sqrt{[Nr]}} \sum_{t=1}^{[Nr]} Y_t. \quad (\text{A.12})$$

From (A.10), we have $\frac{1}{\sqrt{[Nr]}} \sum_{t=1}^{[Nr]} Y_t \xrightarrow{D} \mathcal{N}(0, \sigma^2)$. Additionally, $\frac{\sqrt{[Nr]}}{\sqrt{N}} \rightarrow \sqrt{r}$. Therefore, (A.12) converges in distribution to:

$$\sqrt{N}Y_N(r) \xrightarrow{D} \sqrt{r} \mathcal{N}(0, \sigma^2) = \mathcal{N}(0, r\sigma^2), \quad (\text{A.13})$$

From this, it follows that

$$\frac{\sqrt{N}Y_N(r)}{\sigma} \xrightarrow{D} \mathcal{N}(0, r). \quad (\text{A.14})$$

Moreover, if we consider averages based on observations from $[Nr_1]$ to $[Nr_2]$, with $r_1 < r_2$, we have

$$\sqrt{N} \left[\frac{Y_N(r_2) - Y_N(r_1)}{\sigma} \right] \xrightarrow{D} \mathcal{N}(0, r_2 - r_1). \quad (\text{A.15})$$

We conclude from (A.14) and (A.15) that the sequence of random functions $\left\{ \frac{\sqrt{N}Y_N(\cdot)}{\sigma}, N = 1, 2, \dots \right\}$ has a limiting distribution which is the **Standard Brownian Motion**:

$$\frac{\sqrt{N}Y_N(\cdot)}{\sigma} \xrightarrow{D} W(\cdot). \quad (\text{A.16})$$

Equivalently, for each $0 \leq r \leq 1$, the random variable $\frac{\sqrt{N}Y_N(r)}{\sigma}$ has as its limiting distribution $\mathcal{N}(0, r)$, as given in (A.14). If $r = 1$, we have $Y_N(1) = \frac{1}{N} \sum_{t=1}^N Y_t$, and we obtain as a result the **usual CLT**, the limiting distribution being $\mathcal{N}(0, 1)$.

The **Continuous Mapping Theorem** states that, if $X_N \xrightarrow{D} X$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function, then

$$g(X_N) \xrightarrow{D} g(X). \quad (\text{A.17})$$

In the case of stochastic processes, this result can be generalized by now considering $g(\cdot)$ as a continuous functional. For this purpose, we need to modify the definition of **convergence in distribution for stochastic processes**.

Definition A.8. We say that $S_N(\cdot) \xrightarrow{D} S(\cdot)$ if:

(i) For any $0 \leq r_1 \leq \dots \leq r_k \leq 1$,

$$y_N = \begin{bmatrix} S_N(r_1) \\ \vdots \\ S_N(r_k) \end{bmatrix} \xrightarrow{D} y = \begin{bmatrix} S(r_1) \\ \vdots \\ S(r_k) \end{bmatrix}.$$

(ii) For every $\varepsilon > 0$, $P\{|S_N(r_1) - S_N(r_2)| > \varepsilon\} \rightarrow 0$, uniformly in N , for all r_1, r_2 such that $|r_1 - r_2| < \delta$, $\delta \rightarrow 0$.

(iii) $P\{|S_N(0)| > \lambda\} \rightarrow 0$, uniformly in N , as $\lambda \rightarrow \infty$.

Under these conditions, if $S_N(\cdot) \xrightarrow{D} S(\cdot)$ and g is a continuous functional, then

$$g(S_N(\cdot)) \xrightarrow{D} g(S(\cdot)). \quad (\text{A.18})$$

For example, we saw that $\sqrt{N}Y_N(r) \xrightarrow{D} \sigma W(r)$. If we consider $S_N(r) = [\sqrt{N}Y_N(r)]^2$, then

$$S_N(\cdot) \xrightarrow{D} \sigma^2 [W(\cdot)]^2. \quad (\text{A.19})$$

A.4 Proof Theorem 1.1

From (1.32) and denoting $\hat{\phi} = \phi_{OLS}$, we may write:

$$T(\hat{\phi} - 1) = \frac{\frac{1}{T} \sum_{t=1}^T Y_{t-1} \varepsilon_t}{\frac{1}{T^2} \sum_{t=1}^T Y_{t-1}^2}. \quad (\text{A.20})$$

If $\phi = 1$ and $Y_0 = 0$, we have $Y_t = \sum_{j=1}^t \varepsilon_j$, and $Y_t \sim N(0, \sigma^2 t)$. Also,

$$\begin{aligned} Y_t^2 &= (Y_{t-1} + \varepsilon_t)^2 = Y_{t-1}^2 + \varepsilon_t^2 + 2Y_{t-1}\varepsilon_t, \\ &\implies \\ Y_{t-1}\varepsilon_t &= \frac{1}{2}(Y_t^2 - Y_{t-1}^2 - \varepsilon_t^2) \end{aligned}$$

from which we obtain

$$\begin{aligned} \sum_{t=1}^T Y_{t-1}\varepsilon_t &= \frac{1}{2} \sum_{t=1}^T (Y_t^2 - Y_{t-1}^2 - \varepsilon_t^2) = \frac{1}{2}(Y_T^2 - Y_0^2) - \frac{1}{2} \sum_{t=1}^T \varepsilon_t^2 \\ &= \frac{1}{2}Y_T^2 - \frac{1}{2} \sum_{t=1}^T \varepsilon_t^2. \end{aligned}$$

Dividing both sides by $T\sigma^2$ gives

$$\frac{1}{T\sigma^2} \sum_{t=1}^T Y_{t-1}\varepsilon_t = \frac{1}{2} \left(\frac{Y_T}{\sigma\sqrt{T}} \right)^2 - \frac{1}{2\sigma^2 T} \sum_{t=1}^T \varepsilon_t^2. \quad (\text{A.21})$$

Since $Y_T/(\sigma\sqrt{T}) \sim N(0, 1)$, it follows that $(Y_T/(\sigma\sqrt{T}))^2 \sim \chi^2(1)$. Moreover, since ε_t are i.i.d. random variables with $E(\varepsilon_t^2) = \sigma^2$, by the law of large numbers we have

$$\frac{1}{T} \sum_{t=1}^T \varepsilon_t^2 \xrightarrow{p} \sigma^2. \quad (\text{A.22})$$

Combining (A.21) and (A.22), we conclude that

$$\frac{1}{\sigma^2 T} \sum_{t=1}^T Y_{t-1} \varepsilon_t \xrightarrow{\mathcal{D}} \frac{1}{2}(X - 1), \quad (\text{A.23})$$

where $X \sim \chi^2(1)$. Let us now analyze the denominator of (A.20). Since ε_t are i.i.d. with zero mean and variance σ^2 , we can form the partial sums as in (A.11):

$$Y_T(r) = \frac{1}{T} \sum_{t=1}^{[Tr]} \varepsilon_t,$$

where, from the R.W. formula:

$$Y_T(r) = \begin{cases} 0, & \text{if } 0 \leq r < 1/T, \\ Y_1/T, & \text{if } 1/T \leq r < 2/T, \\ \vdots & \\ Y_T/T, & \text{if } r = 1. \end{cases} \quad (\text{A.24})$$

From (A.13) and (A.19):

$$S_T(r) = [\sqrt{T}Y_T(r)]^2 \xrightarrow{\mathcal{D}} \sigma^2[W(r)]^2, \quad (\text{A.25})$$

where $W(r)$ denotes the standard Brownian motion. Moreover, considering

$$S_T(r) = \begin{cases} 0, & \text{if } 0 \leq r < 1/T, \\ Y_1^2/T, & \text{if } 1/T \leq r < 2/T, \\ \vdots & \\ Y_T^2/T, & \text{if } r = 1, \end{cases} \quad (\text{A.26})$$

it follows that

$$\int_0^1 S_T(r) dr = \frac{1}{T} \sum_{t=1}^T S_T(T) = \frac{1}{T^2} \sum_{t=2}^T Y_{t-1}^2. \quad (\text{A.27})$$

From (A.25) and again by the Continuous Mapping Theorem, we obtain

$$\frac{1}{T^2} \sum_{t=2}^T Y_{t-1}^2 \xrightarrow{\mathcal{D}} \sigma^2 \int_0^1 [W(r)]^2 dr. \quad (\text{A.28})$$

Since $T(\hat{\phi} - 1)$ is a continuous function of (A.23) and (A.28), we obtain equation (1.35), recalling that $W(1)^2 \sim \chi^2(1)$.

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